Week 41: Portals

Adult guide built from the problem files · Unpiloted. Mats, copies and tracing paper untested.

Problems and routes

Problem	K–1	2–3	4–5	route
Predict where four trips from H end	P1	P1	P1	first
Trips from H back to H	P2	P2		first
Trips home with 2, 3, 4 and 5 steps	P3			return
Replay trips on the repeated map		P3		return
Name the finishing copy			P2	first
Four trips home that finish in different copies			P3	first
Can two pawns trade places?	P4			return
Same place, same copy?		P4		return
Which trip shrinks to staying at H?			P4	return
Change RRRUUU into UUURRR			P5	return
Visit another copy and come home		P5		return
When can two trips home be changed into each other?			P6	return

First route. K–1 P1–P2; Grades 2–3 P1–P2; Grades 4–5 P1–P3. **Return visits.** K–1 P3–P4; Grades 2–3 P3–P5; Grades 4–5 P4–P6.

Key, one entry per problem

Predict where four trips from H end

K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1 · first route

Idea. A step across an edge comes back in on the opposite edge at the same height or place across.

K–1: The four printed routes end, from top to bottom, at F, G, F, F. RRD: H → E → D → F. UU: H → B → G. LD: H → D → F. UURR: H → B → G → I → F.

Several different routes intentionally reach F. At a portal continue in the same arrow direction, and a seam crossing still counts as one ordinary step.

Grades 2–3: In printed order: RRD→F, UU→G, LD→F, UURR→F. Three different routes reach F; the path itself is more information than its final portal cell.

Grades 4–5: Destinations: F, G, F, F for RRD, UU, LD, UURR.

Trips from H back to H

K–1 P2 · Grades 2–3 P2 · first route · wording differs by band

Idea. K-1: every cell is exactly two steps from H. Grades 2-3: the three-step trips home are exactly RRR, LLL, UUU and DDD; six steps allow many more.

K–1: Every one of the nine cells is reachable in exactly two steps. A: LU. B: DD. C: RU. D: RR. H: RL. E: LL. F: LD. G: UU. I: RD.

This list is complete because it contains every cell in the map. The witnesses use wrapping: for example DD goes H→G→B, whereas UU goes H→B→G. Exactly two steps does not mean at most two.

Plain-board contrast for Problems 2 and 3: with no portals, two steps from H reach only H, A, C, F and I, and no odd trip returns to H, since each step changes checkerboard colour.

Grades 2–3: Exactly three steps: the complete list is RRR, LLL, UUU, DDD. Exactly six steps: examples include RRRRRR, LLLLLL, UUUUUU, DDDDDD, RRRUUU, RURURU, RRRDDD, LLLUUU, LLLDDD, and RLRLRL.

A six-step word returns to H precisely when its small-step displacement is (0,0), ($\pm$6,0), (0,$\pm$6), or ($\pm$3,$\pm$3). There are 400 balanced strings, four straight six-step strings, and 20 for each of the four diagonal sign choices: 484 total.

Trips home with 2, 3, 4 and 5 steps

K–1 P3 · return route

Idea. A trip home with an odd number of steps must lap the board.

Many answers are valid. Two examples at each requested length: 2 steps: RL and UD. 3 steps: RRR and UUU. 4 steps: RRLL and RULD. 5 steps: RRRUD and UUURL.

Every route listed finishes at H. A candidate is valid exactly when both signed coordinate displacements are multiples of three; adults may use that check without demanding arithmetic from the child.

Replay trips on the repeated map

Grades 2–3 P3 · return route

Idea. Replaying on the repeated map shows which copy a trip finishes in: RRLL returns to the original copy, RRRUUU does not.

The starting H is in the original copy. RRR: one copy right, (1,0). UUU: one copy up, (0,1). RRRUUU: one copy right and up, (1,1). RRLL: original copy, (0,0).

Each route finishes at H on the portal map. The first three do not finish in the original copy. For RRLL, the route crosses into the right copy and comes back; count the signed displacement, not just whether another copy was visited.

Name the finishing copy

Grades 4–5 P2 · first route

Idea. Each R adds a third of a copy to the first coordinate and each U a third to the second; the finishing copy is the net count over three.

In printed order, RRRUUU and UUURRR both finish in (1,1); RRLL and RRRUUULLLDDD both finish in (0,0). Each full copy is three small-cell steps, not one. Add right-minus-left and up-minus-down steps, then divide both by three for a home-returning route. The copy labels on the map give the coordinates.

Four trips home that finish in different copies

Grades 4–5 P3 · first route

Idea. A trip returns to H exactly when both net step counts are multiples of 3; different multiples give different copies of H.

Four examples returning to H in different copies: RRR$\rightarrow$(1,0), LLL$\rightarrow$($-$1,0), UUU$\rightarrow$(0,1), DDD$\rightarrow$(0,$-$1). Many other choices are valid; the copy pair must differ for each of the four.

A follow-up to ask aloud: which copies can a trip home of exactly 4, 5 or 6 steps reach? Four steps: only (0,0). Five steps: the four neighbouring copies ($\pm$1,0) and (0,$\pm$1). Six steps: (0,0), the four diagonal copies ($\pm$1,$\pm$1), and ($\pm$2,0) and (0,$\pm$2).

Can two pawns trade places?

K–1 P4 · return route

Idea. Two pawns given the same steps keep the same offset, so pawns on H and E can never trade places.

No. Every shared arrow sequence translates both pawns equally. The pawn initially at E remains one cell to the right of the pawn initially at H, including across portals. If H moved to E, that translation is one cell right, so E would move to D, not H.

Same place, same copy?

Grades 2–3 P4 · return route

Idea. All three finish at H on the portal map; RRRUUU and RURURU finish in copy (1,1), RRRDDD in (1,-1).

All three trips finish at H on the portal map. RRRUUU and RURURU finish in copy (1,1) on the repeated map; RRRDDD finishes in copy $(1,-1)$, one copy right and one down. The first two drawn paths differ, but both have three right and three up steps. The same place on the portal map need not be the same copy. This question asks for endpoint comparison, not yet a full deformation proof.

Which trip shrinks to staying at H?

Grades 4–5 P4 · return route

Idea. RRRUUULLLDDD finishes in copy (0,0) and shrinks to staying at H; RRR finishes in (1,0) and cannot.

RRRUUULLLDDD can shrink to staying at H; RRR cannot. The first has zero lifted displacement. An explicit legal reduction is to commute each L leftward across each U, using nine square slides, producing RRRLLLUUUDDD. Cancel the three adjacent RL pairs and then the three UD pairs. The second finishes at H in copy (1,0). Every allowed move preserves the lifted endpoint, so it cannot become the empty path at the original H.

Change RRRUUU into UUURRR

Grades 4–5 P5 · return route

Idea. Nine square slides, one per square between the trips, change RRRUUU into UUURRR.

One full nine-slide sequence is: RRRUUU $\to$ RRURUU $\to$ RURRUU $\to$ URRRUU $\to$ URRURU $\to$ URURRU $\to$ UURRRU $\to$ UURRUR $\to$ UURURR $\to$ UUURRR.

Each step swaps one adjacent R,U to U,R through one square of cell centers; its two subpath endpoints stay fixed. The whole route starts in the same copy throughout.

Visit another copy and come home

Grades 2–3 P5 · return route

Idea. A trip can leave the original copy and return to it, visiting a copy above and a copy to the right on the way.

RRRLLL visits the right copy and returns to the original H. To visit a copy above and a copy to the right, use UUUDDDRRRLLL, or RRRUUULLLDDD. The latter passes through right, upper-right, and upper copies before returning to the original. Accept any legal route with zero net small-step displacement meeting the visit condition.

When can two trips home be changed into each other?

Grades 4–5 P6 · return route

Idea. Two trips from H back to H change into each other by backtracks and square slides exactly when they finish in the same copy: loops on the torus are classified by pairs of whole numbers.

Exactly when their finishing copies agree. Necessity: a backtrack has zero displacement and either side of a square slide has the same endpoints in the tiled plane. Hence every allowed move preserves the endpoint copy. Sufficiency: square slides commute adjacent perpendicular steps, so a path can be rearranged into horizontal steps followed by vertical steps. Cancel adjacent opposite horizontal steps, then opposite vertical steps. It becomes the canonical path determined by its signed horizontal and vertical displacement. Equal endpoint copies for H-based routes give equal displacements and therefore the same canonical route. Reverse the second path's reduction, using backtrack insertions where needed, to connect the originals.